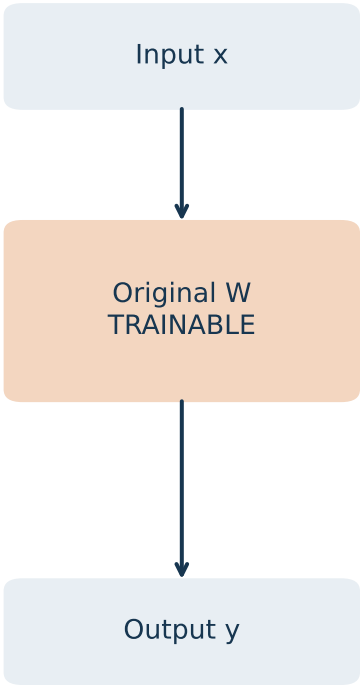
How LoRA changes training

Same pretrained starting weights and the same new training data

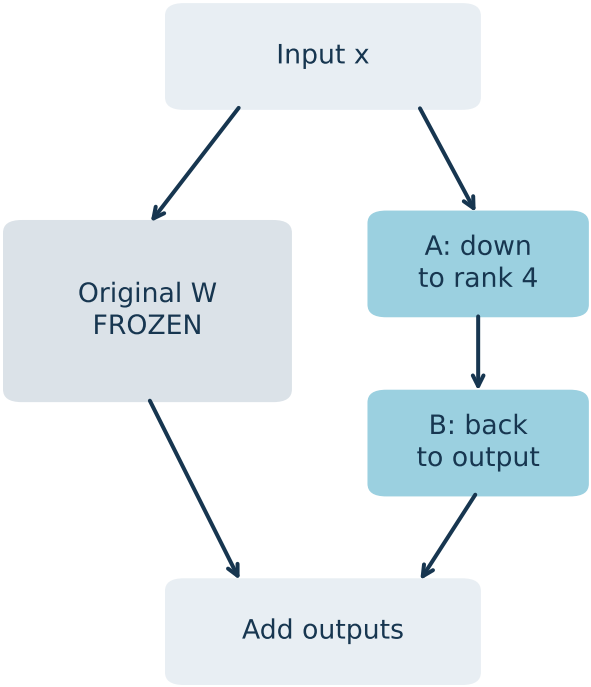
Both approaches learn from our basketball examples. Full fine-tuning lets every original model parameter change. LoRA freezes the original parameters and learns small added matrices. The diagram shows one adapted linear projection; the experiment applies adapters throughout the transformer.

FULL FINE-TUNING



y = Wx

LORA FINE-TUNING



y = Wx + (alpha / rank) BAx

Only the blue A and B matrices train in the LoRA model. Rank = 4; alpha = 8. Biases are omitted from this conceptual diagram. LoRA adapters were applied to QKV, attention-output, and both MLP projections in all 12 transformer blocks.

163,037,184

trainable in full fine-tuning

589,824

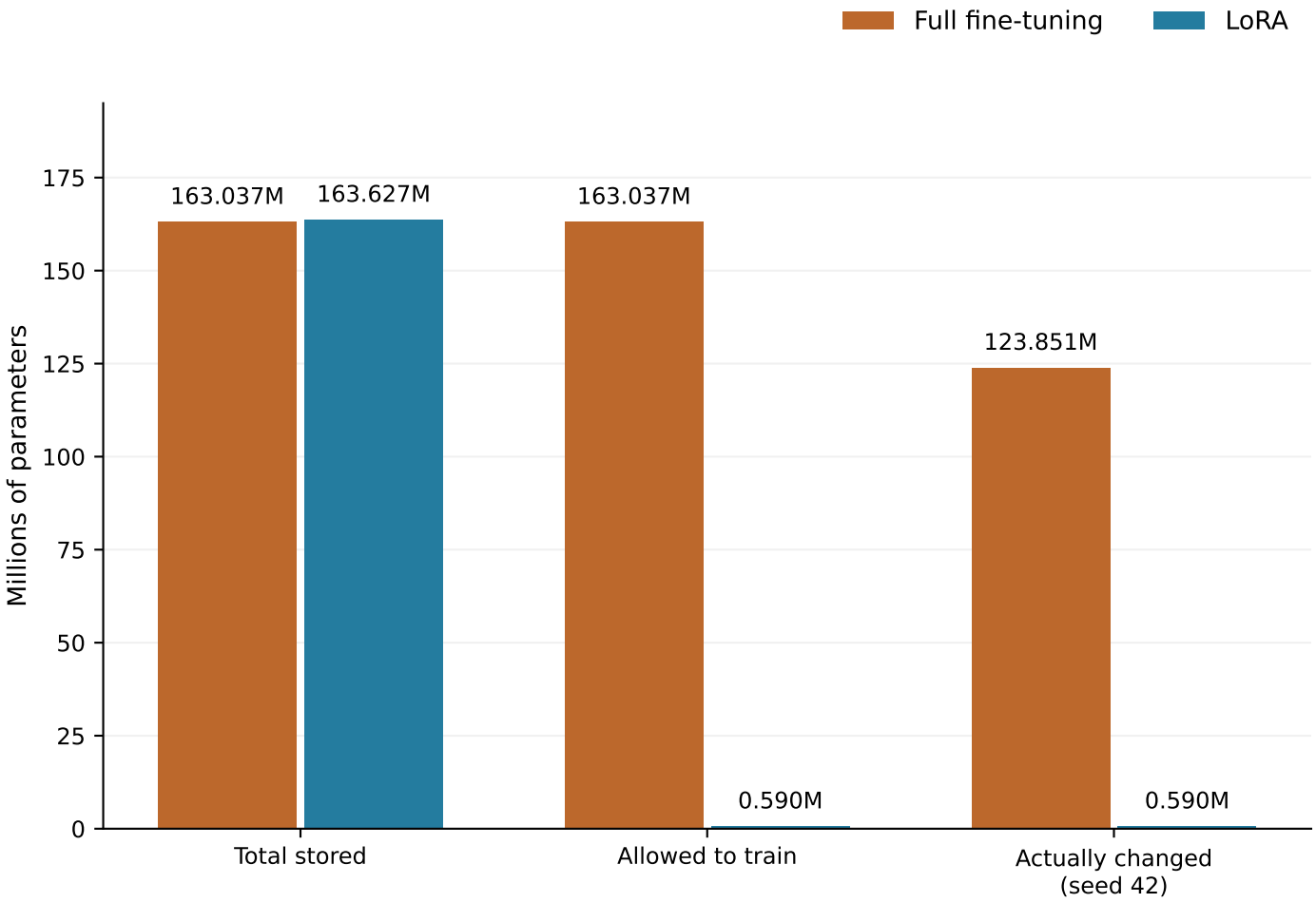
trainable with LoRA

The untouched pretrained model is a third reference: it receives no task training. The two trained models use identical examples, sample order, batch size, and step count. This isolates parameter-efficient adaptation rather than comparing training against no training.

Source: saved matched experiment reports | Conceptual diagrams labeled separately

Total, trainable, and actually changed

These are different quantities; all counts refer to individual scalar values



Total: every model parameter, including frozen weights. Trainable: parameters allowed to receive optimizer updates. Actually changed: individual values differing from their pre-training snapshots at the selected checkpoint. Being trainable does not guarantee a value changed.

276.4x fewer

trainable parameters with LoRA

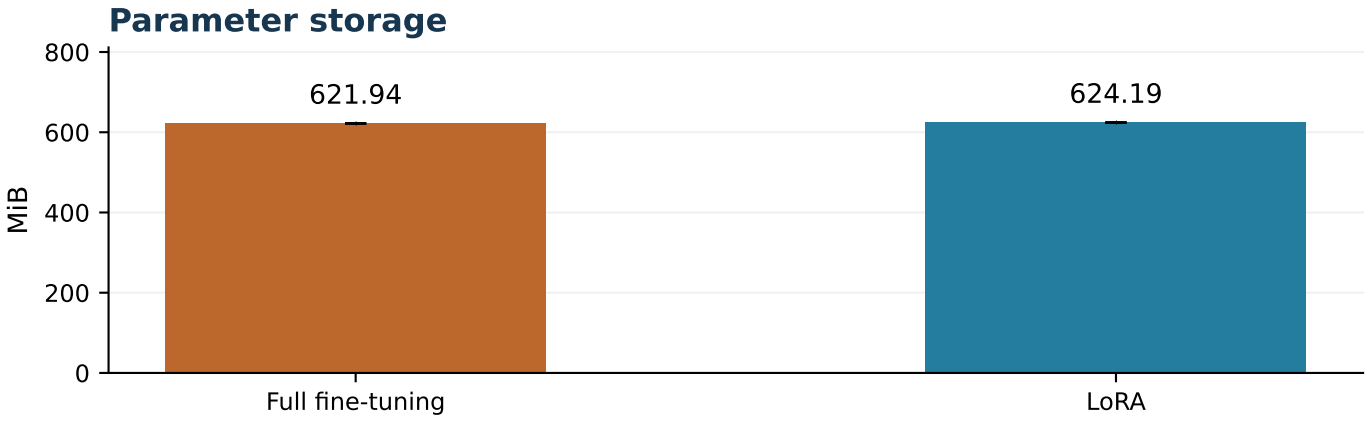
0

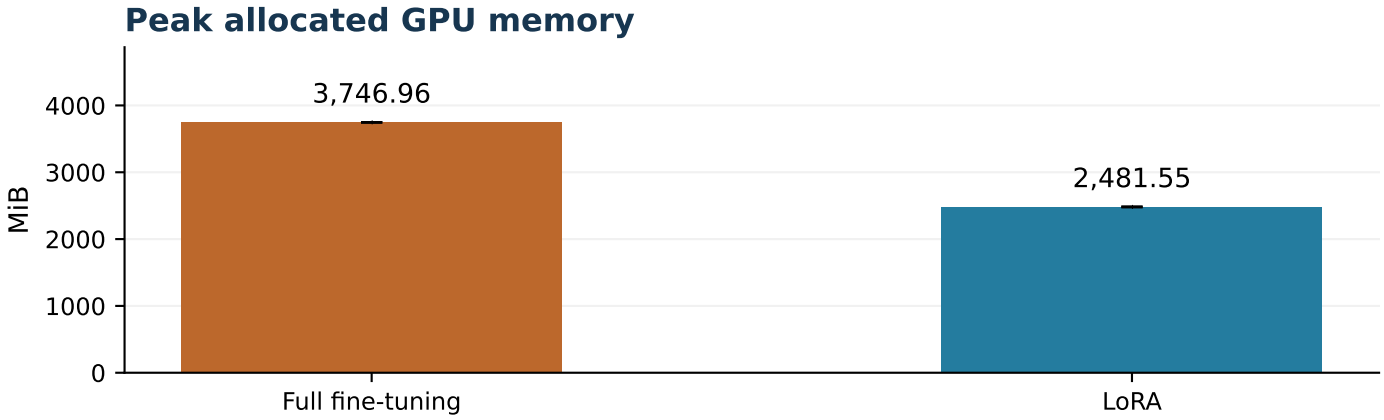
frozen base values changed

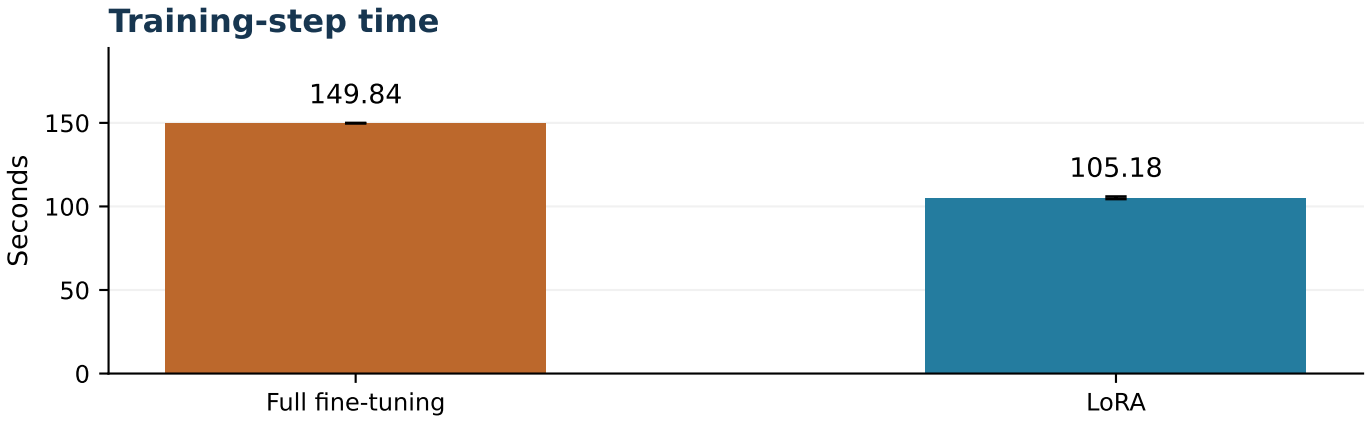
LoRA makes 0.3605% of its total parameters trainable. Every LoRA base-weight audit passed for all three seeds. Full fine-tuning changed 123,851,224 values in seed 42; LoRA changed all 589,824 adapter values. The chart uses a zero baseline; small adapter bars are labeled explicitly.

Training resources: what is saved?

NVIDIA A30 | FP32 | 1,200 steps | batch 8 | sequence length 128



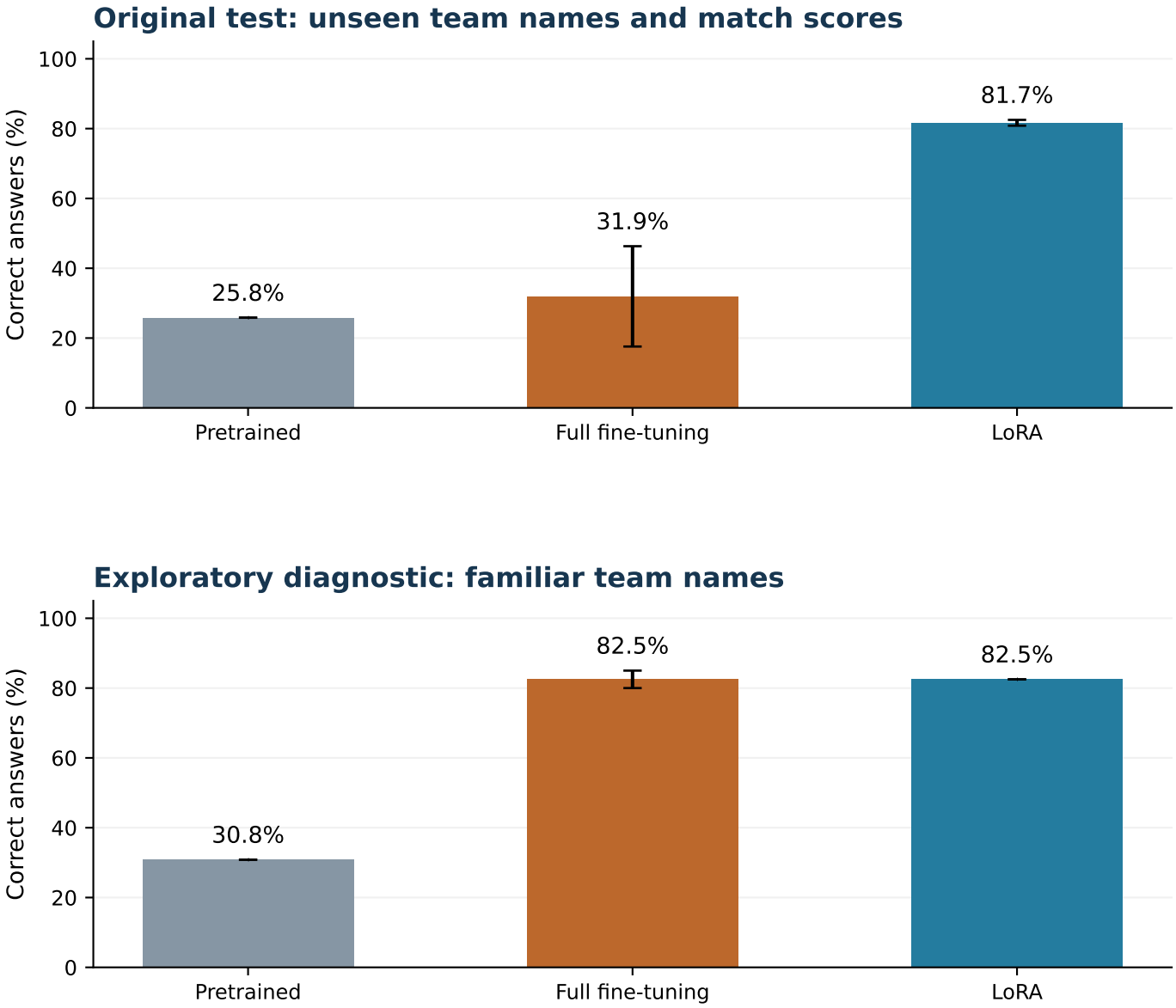




LoRA adds 2.25 MiB of FP32 weights, but reduces measured peak allocated training memory by 33.8% and training-step time by 29.8%. Runtime memory also includes gradients, optimizer states, activations, and temporary allocations. Time excludes transfers, validation, audits, checkpointing, and testing. Bars show three-seed means; error bars show sample standard deviation.

Quality alongside parameter efficiency

Correct primary answers | means and sample standard deviations over three seeds



Original test: 120 questions, with test team names and score combinations absent from training; none of its full reference answers occur in training. LoRA generalized better to unfamiliar names in this experiment. Full fine-tuning often substituted training-team names.

The familiar-name diagnostic was added after inspecting failures. Only names changed; scores and checkpoints stayed fixed. 9 of 120 reference answers overlap training. Its similar accuracy is encouraging but does not establish statistical equivalence.

These are synthetic, format-sensitive question-answering scores, not broad conversational quality. Both methods scored around 50% on comeback questions. Error bars are seed-to-seed variation, not confidence intervals.

Source: saved matched experiment reports | Conceptual diagrams labeled separately